

of life based on the current rules, and we must, therefore, make plans based on the current tax system.

By this point, you should have a good sense of your needs, and those needs should be specified on a pretax basis. In the example we've been using, you have \$200,000 in a tax-sheltered plan and will need to withdraw \$16,000 annually, in after-inflation, pretax terms.

Roughly speaking, therefore, your Needs-to-Wealth (abbreviated as NtW) ratio is $\$16,000/\$200,000 = 8\%$. Another way of looking at it is to say that your annual income needs represent 8% of the initial wealth available to support those needs. Thus, if you had \$400,000 and your needs were 32,000, you would also have an 8% Needs-to-Wealth ratio. This ratio is important because it gives you a general sense of what kind of investment returns you will require to support your annual needs. I would argue that all people—at age 65—with an 8% NtW ratio are more or less in the same boat. That's because whether they have \$1,000,000 or a mere \$100,000 at retirement, they all have the same relative needs.

We are now ready to revisit the main question. Is \$200,000 enough to support \$16,000 in annual needs? The answer, of course, really depends on how you invest the \$200,000. Another way of asking the question is, can you sustain an NtW ratio of 8%? The answer to this question clearly depends on the holy grail of asset allocation. In other words, it depends on what your investment portfolio looks like during your retirement years.

First of all, let's examine the scenario in which you, as a retiree, will live on the interest and dividends alone. In this case, the \$200,000 will have to generate exactly 8% annually to create \$16,000—after inflation and before taxes. Unfortunately, money market instruments will earn nowhere near that amount. So, you have a clear choice: 1) invest in these relatively safe investments, knowing that you eventually will have to liquidate your capital and may run out of money or 2) invest a bit more aggressively, and hopefully build your capital instead. Or, of course, you can always decide to reduce consumption.

The math is relatively simple. If your money earns a fixed 5% in real terms, and you consume \$16,000 in real terms every year, you will run out of money in about 20 years. That's because the present value of \$16,000, discounted at the rate of 5%, is exactly equal to \$200,000.